King Fahd University of Petroleum and Minerals
Department of Mathematics
Student effort not Official. Generated by Ice ❄. دعواتكم له

Math 201 Major Exam I Term 231

Original exam date: 4 October 2023. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

---

1. The area of the region that lies inside the polar curve $r = 2 + \sin\theta$ and outside the circle $r = 3\sin\theta$ is

(a) 3π

(b) $\dfrac{5\pi}{2}$

(c) $\dfrac{9\pi}{4}$

(d) π

(e) $\dfrac{4\pi}{3}$

2. If the equation $x^2 + y^2 + z^2 + 2x - 4y + 6z + 5 = 0$ represents a sphere with center (a, b, c) and radius r, then $a + b + c + r =$

(a) -3

(b) 0

(c) 1

(d) -2

(e) 2

3. A vector $\vec{v}$ of length 3 that has the direction opposite to the vector $\vec{a} = \langle 1, 2, -3\rangle$ is

(a) $\dfrac{1}{\sqrt{11}}\langle 3, 6, -9\rangle$

(b) $\dfrac{1}{\sqrt{11}}\langle 1, 2, -3\rangle$

(c) $\dfrac{1}{\sqrt{14}}\langle -3, -6, 9\rangle$

(d) $\dfrac{1}{\sqrt{14}}\langle 2, 4, -6\rangle$

(e) $\dfrac{1}{\sqrt{14}}\langle -1, -2, 9\rangle$

4. If $\vec{u} = \langle 2, -4\rangle$, $\vec{v} = \langle 2, -3\rangle$ and a, b are scalars such that $a\vec{u} + b\vec{v} = \langle -4, 1\rangle$, then $2a + b =$

(a) 1

(b) -2

(c) 3

(d) 2

(e) 0

5. A set of parametric equations for the rectangular equation $y = 2x - 5$ that represents the point $(3, 1)$ when $t = 0$ is

(a) $x = 3 + t,\ y = 1 - t$

(b) $x = 3 - 2t,\ y = 1 + t$

(c) $x = 3 + 2t,\ y = 1 + 2t$

(d) $x = 3 - t,\ y = 2 + t$

(e) $x = 3 - t,\ y = 1 - 2t$

6. Consider the vectors $\vec{u} = \langle -3, 1, 2\rangle$ and $\vec{v} = \langle 1, 2, -3\rangle$. If the vector projection of $\vec{u}$ onto $\vec{v}$ is $\langle a, b, c\rangle$, then $a + b + c =$

(a) 3

(b) 2

(c) -1

(d) 0

(e) 1

7. The area enclosed by one loop of the four-leaved rose $r = \cos 2\theta$ is equal to

(a) $\dfrac{5\pi}{8}$

(b) $\dfrac{2\pi}{3}$

(c) π

(d) 8π

(e) $\dfrac{\pi}{8}$

8. The slope of the tangent line to the cardioid $r = 1 + \sin\theta$ when $\theta = \dfrac{\pi}{3}$ is equal to

(a) 2

(b) $-\dfrac{1}{3}$

(c) 3

(d) -2

(e) -1

9. The sum of all possible values of m such that the points $(0, 2, 1)$, $(m-1, 0, m)$, $(5, -m, 6)$ are collinear is equal to

(a) 3

(b) 0

(c) -1

(d) 2

(e) -2

10. The Cartesian equation of the parametric curve $x = \ln t$, $y = \sqrt{t}$, $t \geq 1$ is given by

(a) $y = e^{2x},\ x \geq 0$

(b) $y = e^{x},\ x \geq 1$

(c) $y = e^{x/2},\ x \geq 0$

(d) $y = e^{x},\ x \geq 0$

(e) $y = e^{2x},\ x \geq 1$

11. The area of the triangle with vertices $A(0, 2, 2)$, $B(2, 0, -1)$, $C(3, 4, 0)$ is

(a) $\dfrac{5}{4}$

(b) 5

(c) 1

(d) $\dfrac{15}{2}$

(e) $\dfrac{13}{3}$

12. If $\|\vec{u}\| = \sqrt{3}$, $\|\vec{v}\| = 2$ and $\vec{u} \cdot \vec{v} = \sqrt{6}$, the angle θ between the two vectors $\vec{u}$ and $\vec{v}$ is

(a) $\dfrac{5\pi}{4}$

(b) $\dfrac{\pi}{3}$

(c) $\dfrac{\pi}{4}$

(d) $\dfrac{3\pi}{2}$

(e) $\dfrac{\pi}{5}$

13. The parametric curve $x = t^2 + 4t$, $y = 6t^2$ has a vertical tangent line at the point

(a) $(2, -24)$

(b) $(5, 6)$

(c) $(-4, 24)$

(d) $(-3, 6)$

(e) $(0, 0)$

14. Let C be the portion of the parametric curve $x = 3\cos t$ and $y = 3\sin t$ from the point $(3, 0)$ to $\left(\dfrac{3}{2}, \dfrac{3\sqrt{3}}{2}\right)$. The area of the surface obtained by rotating C about the x-axis is

(a) $\dfrac{2}{3}$

(b) $\dfrac{3}{2}$

(c) 9π

(d) $\dfrac{4}{5}\pi$

(e) 3π

15. The exact length of the polar curve $r = 2\cos\theta$, $0 \leq \theta \leq \pi$ is equal to

(a) 3π

(b) $\dfrac{5\pi}{7}$

(c) $\dfrac{3\pi}{2}$

(d) 5π

(e) 2π

16. If the vectors $\vec{a} = \langle 2, 2, -1 \rangle$ and $\vec{b} = \langle 5, -4, m \rangle$ are orthogonal, then $m =$

(a) 0

(b) 2

(c) -1

(d) 3

(e) -3

17. The slope of the tangent line to the curve $x = 2t - 1$, $y = t + t^2$ at the point $(1, 2)$ is

(a) $\frac{3}{2}$

(b) $\frac{1}{3}$

(c) $-\frac{1}{2}$

(d) 0

(e) 1

18. Consider the points $A(2, 1, -1)$, $B(3, 0, 2)$, $C(4, -2, -1)$ and $D(3, m, 0)$. If the volume of the parallelepiped determined by the vectors $\vec{AB}$, $\vec{AC}$, and $\vec{AD}$ is 4, then the sum of all possible values of m is

(a) $\frac{3}{5}$

(b) $-\frac{2}{3}$

(c) $\frac{3}{7}$

(d) $\frac{3}{2}$

(e) $-\frac{4}{3}$

19. The parametric curve $x = t^2$, $y = t^3 - 3t$ is concave upward on the interval

(a) $t \in (-\infty, 0)$

(b) $t \in (-\infty, \infty)$

(c) $t \in (-1, \infty)$

(d) $t \in (-\infty, 1)$

(e) $t \in (0, \infty)$

20. If $\vec{a}$ and $\vec{b}$ are unit vectors in space and the angle between them is $\frac{2\pi}{3}$, then $(\vec{a} \times \vec{b}) \cdot (\vec{b} \times \vec{a})$ is equal to

(a) -3

(b) 0

(c) $\frac{2}{3}$

(d) $-\frac{3}{4}$

(e) $\frac{4}{5}$

Answer Key

Student effort not Official. Generated by Ice ❄. دعواتكم له

Q	Ans	Brief explanation
1	**(c)**	Circle lies inside the limacon: $\frac{9\pi}{2} - \frac{9\pi}{4} = \frac{9\pi}{4}$.
2	**(c)**	Completing squares: center $(-1, 2, -3)$, $r = 3$, so the sum is 1.
3	**(c)**	$\vec{v} = -3\vec{a}/\|\vec{a}\| = \frac{1}{\sqrt{14}}\langle -3, -6, 9\rangle$.
4	**(c)**	$2a + 2b = -4$, $-4a - 3b = 1$ give $a = 5$, $b = -7$, so $2a + b = 3$.
5	**(e)**	Need $y = 2x - 5$ for all t: $x = 3 - t$ gives $2x - 5 = 1 - 2t = y$.
6	**(d)**	$\frac{-7}{14}\langle 1, 2, -3\rangle$, whose components sum to 0.
7	**(e)**	One petal $-\frac{\pi}{4} \le \theta \le \frac{\pi}{4}$: $A = \frac{1}{2}\int \cos^2 2\theta \, d\theta = \frac{\pi}{8}$.
8	**(e)**	$\frac{dy}{dx} = \frac{r' \sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta} = \frac{(1+\sqrt{3})/2}{-(1+\sqrt{3})/2} = -1$.
9	**(c)**	$(m-1)(m+2) = 10 \Rightarrow m^2 + m - 12 = 0$, roots $3, -4$, sum -1.
10	**(c)**	$t = e^x$, $y = e^{x/2}$, and $t \ge 1$ means $x \ge 0$.
11	**(d)**	$\vec{AB} \times \vec{AC} = \langle 10, -5, 10\rangle$, norm 15, area $\frac{15}{2}$.
12	**(c)**	$\cos\theta = \frac{\sqrt{6}}{2\sqrt{3}} = \frac{\sqrt{2}}{2}$, so $\theta = \frac{\pi}{4}$.
13	**(c)**	$x' = 2t + 4 = 0 \Rightarrow t = -2$, $y' = -24 \neq 0$: point $(-4, 24)$.
14	**(c)**	$t \in [0, \pi/3]$, $ds = 3\,dt$: $S = 18\pi \int_0^{\pi/3} \sin t \, dt = 9\pi$.
15	**(e)**	$\sqrt{r^2 + r'^2} = 2$, so $L = \int_0^{\pi} 2\,d\theta = 2\pi$.
16	**(b)**	$\vec{a} \cdot \vec{b} = 2 - m = 0$, so $m = 2$.
17	**(a)**	$t = 1$; $\frac{dy}{dx} = \frac{1+2t}{2} = \frac{3}{2}$.
18	**(b)**	$\det = 6m + 2$, $\|6m + 2\| = 4$ gives $m = \frac{1}{3}, -1$, sum $-\frac{2}{3}$.
19	**(e)**	$\frac{d^2y}{dx^2} = \frac{3(t^2+1)}{4t^3} > 0$ iff $t > 0$.
20	**(d)**	$-\|\vec{a} \times \vec{b}\|^2 = -\sin^2 \frac{2\pi}{3} = -\frac{3}{4}$.